

# Daesol Cho

ROBOTICS RESEARCHER

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## Education

### SNU (Seoul National University)

PH.D. IN MECHANICAL & AEROSPACE ENGINEERING

- Dissertation: "Autonomous Robot Learning with Minimal Intervention."

Seoul, Korea

September. 2021 - February. 2025

### SNU (Seoul National University)

M.S. IN MECHANICAL & AEROSPACE ENGINEERING

- Thesis topic: "Dual-arm Manipulation Using Hierarchical Reinforcement Learning."

Seoul, Korea

September. 2019 - Aug. 2021

### SNU (Seoul National University)

B.S. IN MECHANICAL & AEROSPACE ENGINEERING

- Thesis topic: "Dynamic Obstacle Removal in ORB-SLAM2 via CNN-based Object Detection."

Seoul, Korea

Mar. 2013 - Aug. 2019

## Experience

### Georgia Institute of Technology

POSTDOCTORAL RESEARCHER

- Research topics: "Robotics, Deep Reinforcement Learning, Foundation Models."

Atlanta, USA

September. 2025 - Present

### Artificial Intelligence Institute of Seoul National University (AIIS)

POSTDOCTORAL RESEARCHER

- Research topics: "Robotics, Deep Reinforcement Learning, Foundation Models."

Seoul, Korea

March. 2025 - August. 2025

## Ongoing works

**Humanoid:** 1) Humanoid VLA with active search & spatial memory: develops a spatial-memory-augmented humanoid VLA for loco-manipulation under partial observability, enabling persistent object-centric scene memory and active re-localization of task-relevant targets beyond the current field of view. 2) Other ongoing work includes physics-aware real-to-simulation for humanoid loco-manipulation, selective human-to-humanoid motion retargeting from human videos, and agentic learning-based safe motion generation for whole-body loco-manipulation.

**Dexterous Hand:** Ongoing work includes guidance-based dexterous hand manipulation, aerial dexterous manipulation, and blind manipulation using tactile and contact sensing.

**Embodied Foundation Models:** Ongoing work includes human-data-driven post-training of vision-language-action models and world-action-model-based generation of robot manipulation data.

## Publication

Morgan Byrd, DongHoon Baek, Kartik Garg, Hyunyoung Jung, **Daesol Cho**, Maks Sorokin, Robert Wright and Sehoon Ha. (2027). AdaptManip: Learning Adaptive Whole-Body Object Lifting and Delivery with Online Recurrent State Estimation, International Conference on Robotics & Automation (ICRA submitted).

Sungha Kim, Gawon Lee, Jusuk Lee, Jonghae Park, H. Jin. Kim and **Daesol Cho**. (2026). FLAG: Flow-based policy MaxEnt-RL by Latent Augmented Guidance, Neural Information Processing Systems (NeurIPS).

Jusuk Lee, Seungjae Lee, Jonghun Shin, Hoseong Jung, Sungha Kim, **Daesol Cho**, H. Jin Kim, Jia-Bin Huang, and Furong Huang (2026). DynaFLIP: Rethinking Robotics Perception via Tri-Modal-Dynamics Guided Representation, Neural Information Processing Systems (NeurIPS).

**Daesol Cho**, Youngseok Jang, Danfei Xu and Sehoon Ha. (2026). [presented in ICRA 2027] EgoAVFlow: Robot Policy Learning with Active Vision from Human Egocentric Videos via 3D Flow, IEEE Robotics and Automation Letters (RA-L).

Jusuk Lee, **Daesol Cho**, Jonghun Shin, Seungyeon Yoo, Jonghae Park, Taekbeom Lee and H. Jin Kim. (2026). BooST: Bridging Semantics and Motions for Efficient Skill Transfer (RA-L).

Hoseong Jung, Sungil Son, **Daesol Cho**, Jonghae Park, Changhyun Choi and H. Jin. Kim. (2026). Temporal Action Representation Learning for Tactical Resource Control and Subsequent Maneuver Generation, International Conference on Robotics & Automation (ICRA).

Jonghae Park, **Daesol Cho**, Jusuk Lee, Dongseok Shim, Inkyu Jang and H. Jin. Kim. (2025). Periodic Skill Discovery, Neural Information Processing Systems (NeurIPS).

**Daesol Cho\***, Seungyeon Yoo\*, Dongseok Shim and H. Jin. Kim. (2025). [presented in ICRA 2026] Single-View 3D-Aware Representations for Reinforcement Learning by Cross-View Neural Radiance Fields, IEEE Robotics and Automation Letters (RA-L).

Jusuk Lee, **Daesol Cho**, Jonghun Shin, Taekbeom Lee, Jonghae Park and H. Jin. Kim. (2025). Unifying What and How: Distilling a Pre-trained Unified Skill Representation for Efficient Adaptation, Conference on Robot Learning (CoRL) workshop.

Gawon Lee, **Daesol Cho** and H. Jin. Kim. (2025). Leveraging Temporally Extended Behavior Sharing for Multi-task Reinforcement Learning, International Conference on Intelligent Robots and Systems (IROS).

Hoseong Jung, Sungil Son, **Daesol Cho**, Jonghae Park, Changhyun Choi and H. Jin. Kim. (2025). Temporal Action Representation Learning for Aerial Maneuvering and Resource-Aware Decision-Making, Robotics: Science and Systems (RSS) workshop.

**Daesol Cho**, Jigang Kim and H. Jin. Kim. (2024). Boosting Autonomous Reinforcement Learning via Action-Free Video and Plasticity Preservation, Robotics: Science and Systems (RSS) workshop.

**Daesol Cho**, Seungjae Lee and H. Jin. Kim. (2023). Diversify & Conquer: Outcome-directed Curriculum RL via Out-of-Distribution Disagreement, Neural Information Processing Systems (NeurIPS).

Seungjae Lee, **Daesol Cho**, Jonghae Park and H. Jin. Kim. (2023). CQM: Curriculum Reinforcement Learning with a Quantized World Model, Neural Information Processing Systems (NeurIPS).

Jigang Kim\*, **Daesol Cho\*** and H. Jin. Kim. (2023). Demonstration-free Autonomous Reinforcement Learning via Implicit and Bidirectional Curriculum, International Conference on Machine Learning (ICML), IROS 2023 workshop

Seungjae Lee, Jongho Shin, Hyeong-Geun Kim, **Daesol Cho** and H. Jin. Kim. (2023). Deep End-to-end Imitation Learning for Missile Guidance With Infrared Images, International Journal of Control, Automation and Systems (IJCAS).

**Daesol Cho\***, Seungjae Lee\* and H. Jin. Kim. (2023). [Spotlight] Outcome-Directed Reinforcement Learning by Uncertainty & Temporal Distance-Aware Curriculum Goal Generation, International Conference on Learning Representations (ICLR).

**Daesol Cho\***, Dongseok Shim\* and H. Jin. Kim. (2022). S2P: State-conditioned Image Synthesis for Data Augmentation in Offline Reinforcement Learning, Neural Information Processing Systems (NeurIPS).

Jigang Kim, J. hyeon Park, **Daesol Cho** and H. Jin. Kim. (2022). [presented in ICRA 2023] Automating Reinforcement Learning With Example-Based Resets, IEEE Robotics and Automation Letters (RA-L).

**Daesol Cho**, Jigang Kim and H. Jin. Kim. (2022). [presented in IROS 2022] Unsupervised Reinforcement Learning for Transferable Manipulation Skill Discovery, IEEE Robotics and Automation Letters (RA-L).

## Projects

### Mobile Humanoid Research for Medical Assistance

PROJECT LEADER FOR ROBOT MANIPULATION

Korea Institute for Advancement of Technology, Georgia Tech, Neuromeka, KAIST

Atlanta, USA

September. 2025 - Present

### Transfer of Driving Dynamics Parameter between Car Models

PROJECT LEADER

Hyundai Motor Company, SNU

Seoul, Korea

April. 2022 - August. 2025

### Transfer Learning for Multi-agent Systems

RESEARCHER

Agency for Defense Development, SNU

Seoul, Korea

October. 2019 - October. 2021

### BabyMind: Infant-Mimic Developmental Machine Learning

RESEARCHER

Korea Ministry of Science and ICT, SNU

Seoul, Korea

April. 2019 - December. 2020

### RL Application of an A/C Unit via Domain Randomization

RESEARCHER

LG Electronics, SNU

Seoul, Korea

August. 2019 - November. 2020

## Honors & Awards & Scholarships

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| 2025      | National Research Foundation of Korea, Sejong Science Fellowship |
| 2023      | Youlchon AI Young Researcher Fellowship                          |
| 2022-2023 | Brain Korea 21 Plus (BK21+) Ph.D Fellowship Scholarship          |
| 2022      | Lecture & Research Scholarship                                   |
| 2019      | Summa Cum Laude, Seoul National University                       |

## Academic Activities

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| 2026      | Invited talk (SNU, Georgia Tech, ETRI, KAIST, SKKU)                                                           |
| 2023-2026 | Reviewer (NeurIPS, ICML, ICLR, ICRA, RA-L, IROS, RSS).                                                        |
| 2024-2025 | Seoul AI Hub, SNU, AI+Robotics Training Program for Skilled Professionals, Reinforcement Learning Instructor. |
| 2023-2024 | Hyundai Motors and LG Group AI Boosting Camp (AIBC) Reinforcement Learning Instructor.                        |